

What Makes Bluetooth Smart

The Internet of Things (IoT) is believed to be an enabler of some impressive technological advances that we are likely to experience in our day-to-day lives in the future. But what makes IoT communicate with other devices in the room without draining out the battery? It is Bluetooth Low Energy



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Developed two decades ago by Ericsson, Bluetooth is one of the most common wireless communication technologies seen in consumer electronics. Unlike Wi-Fi (IEEE 802.11 standards), it usually serves a category of applications known as wireless personal-area network (WPAN). The low energy profile of Bluetooth has made it popular in consumer electronics arena, while Wi-Fi is seen in more institutional setups.

Bluetooth Smart, aka Bluetooth Low Energy

The most important feature of Bluetooth Smart is its extremely low power requirement. Along with its reduced size and very low cost, this allows it to be used on sensors as miniature data loggers. Furthermore, as Bluetooth Smart uses the same frequency band as classic Bluetooth, it is compatible with almost every Bluetooth-enabled mobile device.

Bluetooth Silicon

There are a wide variety of chips available to get you started on integrating Bluetooth Smart technology into your device.

Navaneethan Sundaramoorthy, an independent technology, innovation and education consultant, explains, "There are three broad approaches for integrating Bluetooth into your embedded system design, which vary based on the cost, level of integration and development effort. One option would be to use individual components like an antenna, microcontroller and power supplies directly on your board. The second option is use of pre-integrated Bluetooth PCB modules that can be soldered on your board. Alternatively, you can use off-the-shelf USB (or SDIO) Bluetooth dongles connected to your board."

"Of these three approaches, use of

Bluetooth modules is the most popular approach for embedded systems. These modules have the complete Bluetooth software stack burnt into the microcontroller's Flash memory, and are 100 per cent factory tested and even certified for various wireless standards," he adds.

Single-chip processor. Single-chip processor is the first choice when a designer is looking for the smallest possible footprint along with minimal power consumption, while keeping performance on par with standards. It is also selected when the developer wants to keep the costs as low as possible while implementing his Bluetooth system—for example, when creating Bluetooth client devices.

CSR has a single-chip radio with an integrated 16-bit microprocessor and 64 kB each of RAM and ROM. Named CSR1011, it comes with I²C peripheral and SPI debug interfaces, as well as a development kit and SDK that can be accessed at the CSR website.

Broadcom's BCM20702 is another single-chip Bluetooth processor, but built on the 65nm LP CMOS technol-

Why Bluetooth Smart

Bluetooth Smart finds applications in many consumer electronics, automotive, and industrial, segments such as smartphones and tablets, keyboards and mice, TV remote controls, sports and fitness, toys, gaming, automotive infotainment, etc.

Products which previously used on other proprietary wireless (ISM band) and infra-red (e.g., IrDA) connectivity technologies are migrating to Bluetooth Smart connectivity to benefit from lower power/voltage operation, much longer battery life (years not months), non line-of-sight and two-way operation, etc. The major OS providers have announced the integration of a native Bluetooth Smart stack into their updates and upgrades so that cool "Appcessories" (e.g., a smartphone app for a Bluetooth connected accessory) can be developed. There's an exciting range of upcoming consumer electronics products that are ready for Bluetooth Smart, such as smart watches, door-key fobs, camera accessories, health and fitness monitors, proximity alarms, keyboards, mice, gesture-based remote-controls, tablets and laptops, to name a few.

—Nitin Gupta, Lead Engineer, *µEnergy Applications Group, CSR*